Practice workbook · Quadratic trig equations using an identity

Using sin^2 + cos^2 = 1, a double-angle formula or a proven identity to turn an equation into a quadratic (or polynomial) in one trig ratio, then solving.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-27

1. Question 10 · part c · 4 marks

2021 JUN · Paper 02

Worked-example retry · Attempt part c only.

10 (a) Solve the equation

$$\tan x^\circ = -3 \qquad \text{for } 0 \leqslant x < 360$$

Give your solutions to the nearest whole number.

(3)

Given that

$$7\sin^2\theta + \sin\theta\cos\theta = 6$$

(b) show that

$$\tan^2\theta + \tan\theta - 6 = 0$$

(3)

(c) Hence solve the equation

$$7\sin^2 y^\circ + \sin y^\circ \cos y^\circ = 6 \qquad \text{for } 0 \leqslant y < 360$$

Give your solutions to the nearest whole number.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 10 · part c · 4 marks

2021 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 10 · part c · 4 marks

2021 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 10 · part c · 4 marks

2021 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 10 is 10 marks)

1. Question 10 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 11 · part b · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

11 (a) Using a formula from page 2, show that $\cos 2x = 1 - 2\sin^2 x$

(3)

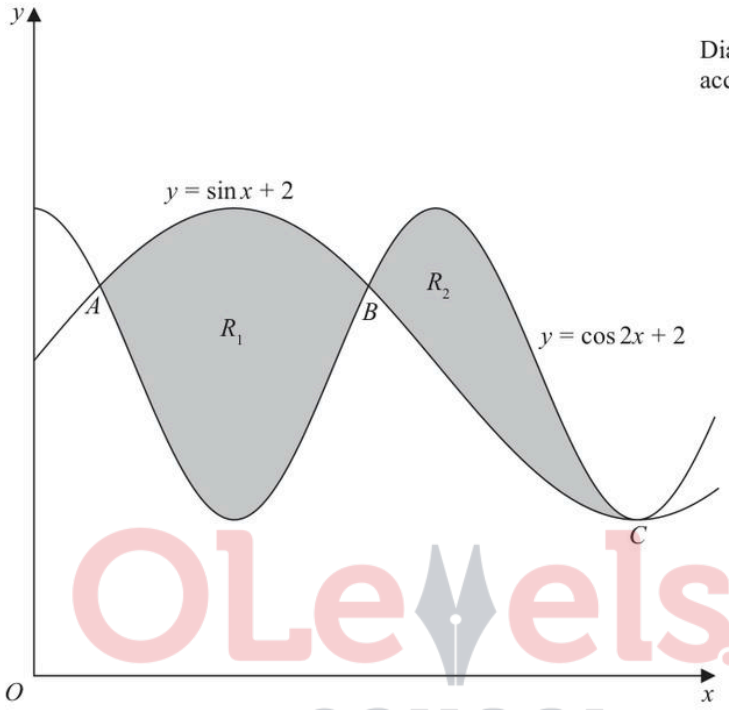


Figure 2

Figure 2 shows a sketch of part of the curves with equations $y = \sin x + 2$ and $y = \cos 2x + 2$

The points A, B and C, shown in Figure 2, are three points that are common to both curves.

(b) Find the coordinates of each of these points.

(4)

R_1 and R_2, shown shaded in Figure 2, are two regions enclosed by the two curves.

(c) Use calculus to find, in its simplest form, the ratio

$$\text{area of } R_1 : \text{area of } R_2$$

(8)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 6 0 2 4 A 0 3 2 3 6

2. Question 11 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 10 · part b · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

10 (a) Using a formula from page 2, show that

(i) $\sin^2 A = \frac{1-\cos 2A}{2}$

(ii) $\cos^2 A = \frac{\cos 2A+1}{2}$

(5)

Diagram **NOT** accurately drawn

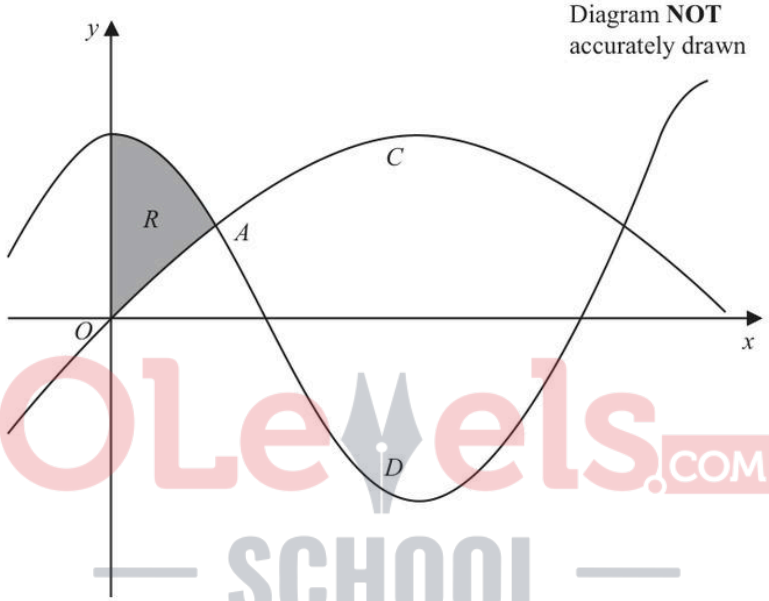


Figure 2

Figure 2 shows part of the curve C with equation $y = \sin x$ and part of the curve D with equation $y = \cos 2x$
Curve C and curve D intersect at the point A

(b) Use algebra to find the exact coordinates of A

(5)

The shaded region R is rotated through $360°$ about the x-axis.

(c) Use algebraic integration to find the exact volume of the solid generated.

Give your answer in the form $\frac{a\sqrt{a}}{b}\pi$

where a is a prime number and b is an integer.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 6 5 0 7 A 0 2 6 3 2

3. Question 10 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 11 · part b · 7 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

11 (a) Show that $\cos^4\theta - \sin^4\theta = \cos 2\theta$

(4)

(b) Hence, or otherwise, solve the equation

$$8\cos^2\left(2\theta + \frac{\pi}{4}\right) - 3 = 2\cos^4\left(\theta + \frac{\pi}{8}\right) - 2\sin^4\left(\theta + \frac{\pi}{8}\right) \quad \text{for} \quad 0 \leqslant \theta < \pi$$

Give your solutions to 2 decimal places.

(7)

(c) Using calculus, find the exact value of $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \left(\cos^4 2x - \sin^4 2x - 8\sin 4x\right) \mathrm{d}x$

Give your answer in the form $a - b\sqrt{2}$ where a and b are rational numbers.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 11 · part b · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 11 · part b · 7 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

11 (a) Using a formula on page 2, show that

$$\frac{1-\cos 2A}{1+\cos 2A} = \tan^2 A$$

(3)

(b) Hence, or otherwise, solve in degrees to one decimal place

$$\frac{3-3\cos 4x}{1+\cos 4x} + \frac{5\sin 2x}{\cos 2x} = 2 \quad \text{for } -90^\circ < x < 90^\circ$$

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 11 · part b · 7 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

Question 11 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 11 · part b · 7 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

Question 11 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 11 is 10 marks)

5. Question 11 · part b · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 11 · part c · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$

(3)

(ii) $\sin 2A = 2\sin A \cos A$

(1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$

(4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$

(4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$

(4)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6. Question 11 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part b · 5 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

8 (i) (a) Using a formula given on page 2, show that

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

(2)

(b) Hence, solve the equation

$$\tan A° - \tan 2A° = 0 \qquad \text{for } 0 \leqslant A \leqslant 180$$

(5)

(ii) Using a formula given on page 2, solve, giving your solutions as exact values

$$\cos\left(x - \frac{\pi}{6}\right) = \sin x \qquad \text{for } -\pi \leqslant x \leqslant 2\pi$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 8 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 7 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

7 (a) Solve the equation

$$\cos(3x - 15)° = \frac{\sqrt{3}}{2} \quad \text{for} \quad 0 \leqslant x < 180$$

(4)

(b) Solve, giving your solutions to one decimal place where appropriate,

$$3\tan y° + 4\sin y° = 0 \quad \text{for} \quad -180 \leqslant y < 180$$

(4)

(c) Solve, giving your solutions to one decimal place where appropriate,

$$\cos\theta° = 3\sin^2\theta° - 1 \quad \text{for} \quad -180 \leqslant \theta < 180$$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

8. Question 7 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 7 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 7 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.



8. Question 7 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 10 · part c · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

10 (a) Solve the equation

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} \qquad \text{for } 0 \leqslant x \leqslant 2\pi$$

Give your solutions in terms of π, where appropriate.

(3)

(b) Solve the equation

$$3\sin\theta + 5\cos\theta = 0 \qquad \text{for } -360° \leqslant \theta \leqslant 360°$$

Give your solutions to the nearest degree.

(3)

(c) Solve the equation

$$1 + \sin 2y = 2\cos^2 2y \qquad \text{for } -180° \leqslant y \leqslant 0°$$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 10 · part c · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 10 · part c · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

32

P 6 2 2 8 3 A 0 3 2 4 0

9. Question 10 · part c · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 10 is 11 marks)

9. Question 10 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 10 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

10 (a) Solve the equation

$$\tan x^\circ = -3 \qquad \text{for } 0 \leqslant x < 360$$

Give your solutions to the nearest whole number.

(3)

Given that

$$7\sin^2\theta + \sin\theta\cos\theta = 6$$

(b) show that

$$\tan^2\theta + \tan\theta - 6 = 0$$

(3)

(c) Hence solve the equation

$$7\sin^2 y^\circ + \sin y^\circ \cos y^\circ = 6 \qquad \text{for } 0 \leqslant y < 360$$

Give your solutions to the nearest whole number.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
OLevels.com SCHOOL

10. Question 10 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 8 · part c · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

8 (a) Solve, to the nearest integer, the equation

$$\sin(2x - 75)° = -0.515 \quad \text{for } 0 \leqslant x < 180$$

(3)

(b) Giving your solutions to one decimal place, where appropriate, solve the equation

$$2\tan y° + 5\sin y° = 0 \quad \text{for } 0 \leqslant y \leqslant 180$$

(4)

(c) Explain mathematically why there are no values of θ that satisfy the equation

$$3\cos^2\theta° - 3\sin^2\theta° + \sin\theta° + 12 = 0$$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 8 · part c · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 8 · part c · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 8 · part c · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 8 is 11 marks)

11. Question 8 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL